

Mock Interview (2 Hours) – Software Engineer Intern @ GreenNode

Role-play: Bạn là ứng viên, người phỏng vấn là kỹ sư tại GreenNode (thuộc VNG – môi trường Cloud, Distributed Systems, AI Infra).

Mục tiêu: Không chỉ trả lời đúng, mà phải thể hiện **engineering mindset, learning mindset, problem solving & collaboration**.

STRUCTURE (2 HOURS)

Section	Time	Nội dung
1	15m	Ice-breaking + Culture fit
2	25m	Programming fundamentals + OOP
3	30m	Backend & System thinking
4	25m	Problem solving & Debugging
5	15m	Git + Workflow + Collaboration
6	10m	Learning mindset & Growth
7	10m	Candidate questions + Wrap-up

1. ICE-BREAKING & CULTURE FIT (15m)

Q1. Introduce yourself

-  **Câu hỏi:** Hãy giới thiệu về bản thân trong 2–3 phút
-  **Nhà tuyển dụng muốn gì:**
 - Khả năng communicate rõ ràng
 - Có định hướng backend/cloud hay không
-  **Kỳ vọng:**
 - Nói ngắn gọn: background → skills → projects → goal
 - Có mention backend / system / cloud nếu có

Q2. Why GreenNode?

-  Vì sao bạn muốn làm ở GreenNode?
-  **Verify:**
 - Có research về công ty không
 - Có hiểu cloud platform là gì không
-  **Expect:**
 - Nhắc tới: cloud, infra, distributed system
 - Muốn học từ production system lớn

Q3. Tell me about a challenge you faced

-  Một vấn đề khó bạn từng gặp và cách giải quyết?
-  **Verify:**
 - Problem-solving mindset
 - Không bỏ cuộc

-  Expect:
 - Có structure: problem → approach → result → lesson

Q4. How do you handle not understanding something?

-  Verify:
 - Learning mindset
 - Có biết hỏi đúng cách không
-  Expect:
 - Try → search → debug → ask đúng người → document lại

2. PROGRAMMING FUNDAMENTALS & OOP (25m)

Q5. Explain OOP concepts

-  OOP là gì? 4 nguyên lý?
-  Verify:
 - Foundation CS
-  Expect:
 - Encapsulation, Inheritance, Polymorphism, Abstraction
 - Có ví dụ thực tế

Q6. Difference between abstract class vs interface?

-  Verify:
 - Hiểu design
-  Expect:
 - Interface = contract
 - Abstract = base class có logic

Q7. What is time complexity?

-  Big-O là gì?
-  Verify:
 - Algorithm thinking
-  Expect:
 - $O(1)$, $O(n)$, $O(\log n)$
 - So sánh loop vs binary search

Q8. Common data structures?

-  Verify:
 - Kiến thức cơ bản
-  Expect:
 - Array, LinkedList, Stack, Queue, HashMap
 - Khi nào dùng cái nào

3. BACKEND & SYSTEM THINKING (30m)

Q9. What happens when you type a URL?

- 🌀 Verify:
 - System thinking
- ✅ Expect:
 - DNS → TCP → HTTP → Server → DB → Response

Q10. What is REST API?

- 🌀 Verify:
 - Backend basics
- ✅ Expect:
 - Stateless
 - CRUD mapping (GET, POST, PUT, DELETE)

Q11. Difference between SQL vs NoSQL?

- 🌀 Verify:
 - Data modeling
- ✅ Expect:
 - SQL: structured, ACID
 - NoSQL: flexible, scale

Q12. How to design a simple API for Todo app?

- 🌀 Verify:
 - API thinking
- ✅ Expect:
 - Endpoints: GET /todos, POST /todos
 - Basic schema

Q13. What is scalability?

- 🌀 Verify:
 - Cloud mindset
- ✅ Expect:
 - Scale vertically vs horizontally
 - Load balancing basic idea

Q14. What is a microservice?

- 🌀 Verify:
 - Cloud-native awareness
- ✅ Expect:
 - Small services, independent deploy
 - Tradeoff: complexity

4. 🐛 PROBLEM SOLVING & DEBUGGING (25m)

Q15. Your API is slow. What do you do?

- 🌀 Verify:
 - Debug mindset

-  Expect:
 - Check logs → DB query → network → profiling

Q16. A bug only happens in production

-  Verify:
 - Real-world thinking
-  Expect:
 - Reproduce → logging → compare env → fix safely

Q17. You don't know how to solve a problem

-  Verify:
 - Attitude
-  Expect:
 - Break problem → Google → docs → ask

Q18. Debugging approach?

-  Verify:
 - Systematic thinking
-  Expect:
 - Hypothesis → test → narrow down → fix

5. GIT & WORKFLOW (15m)

Q19. Git workflow you used?

-  Verify:
 - Team experience
-  Expect:
 - feature branch, pull request

Q20. What is merge vs rebase?

-  Verify:
 - Git understanding
-  Expect:
 - merge: keep history
 - rebase: linear history

Q21. Code review experience?

-  Verify:
 - Collaboration
-  Expect:
 - Learn from feedback
 - Improve code quality

6. LEARNING MINDSET & GROWTH (10m)

Q22. How do you learn new technology?

- 🎯 Verify:
 - Self-learning ability
- ✅ Expect:
 - Docs → tutorial → build project

Q23. What do you do when stuck for hours?

- 🎯 Verify:
 - Efficiency
- ✅ Expect:
 - Timebox → ask → avoid wasting time

Q24. Your career goal?

- 🎯 Verify:
 - Alignment
- ✅ Expect:
 - Backend / Cloud / System engineer

7. ? CANDIDATE QUESTIONS (10m)

👉 Ứng viên NÊN hỏi:

Q25. What does success look like for this role?

- 🎯 Shows:
 - Growth mindset

Q26. What kind of systems will I work on?

- 🎯 Shows:
 - Interest in real system

Q27. How does mentorship work here?

- 🎯 Shows:
 - Wants to learn

⚠️ BONUS – RED FLAGS (NHÀ TUYỂN DỤNG SẼ ĐỂ Ý)

- ❌ Không biết debug
- ❌ Không hỏi khi không hiểu
- ❌ Trả lời lý thuyết nhưng không có ví dụ
- ❌ Không biết Git workflow
- ❌ Không có curiosity

💡 FINAL TIPS

- Luôn nói theo structure: **Problem** → **Approach** → **Result**
- Nếu không biết → nói thẳng + cách bạn sẽ tìm hiểu

- Ưu tiên:
 - Clear thinking
 - Honest
 - Willing to learn